Introduction to Multimodal Learning: Problem Set 3

Due: May 31, 2026
This assignment studies video motion and temporal consistency. You will build a small CPU-friendly tool that predicts frames with motion compensation, visualizes motion vectors and residuals, evaluates short clips, and connects these ideas to modern video models.
Please submit a zip file containing:

- `HW3.ipynb` with all code cells executed and visible outputs.
- A `result` folder with visualizations, metric tables, residual maps, motion-vector plots, and GIFs or MP4s when applicable.
- A `report.pdf`, no more than 4 pages, with key figures, metric tables, short answers, and one concise failure-case discussion. Report clarity and reproducibility are worth 5 pts.

Total: 100 pts + 10 bonus pts.
Starter files include `HW3.ipynb` and `hw3_utils.py`. Recommended packages are listed in the notebook.

1 Motion-Compensated Frame Prediction (45 pts)

Use a TA-provided real clip or the synthetic translation clip from `HW3.ipynb`. Convert frames to grayscale, resize to about 128×128, and evaluate at least 5 consecutive frame pairs.

1.1 Previous-Frame Baseline (10 pts)

Implementation Requirements: Implement:

```
def frame_difference(prev_frame, cur_frame):
    ...
```

The baseline predictor is

$$\hat{I}_t = I_{t-1}, \qquad R_t = I_t - \hat{I}_t.$$

Return $\hat{I}_t$ and R_t. Visualize I_{t-1}, I_t, $|I_t - I_{t-1}|$, and the signed residual. Compute MSE, PSNR, and residual entropy:

$$\mathrm{MSE} = \frac{1}{HW}\sum_{x,y}(I_t(x,y) - \hat{I}_t(x,y))^2, \qquad \mathrm{PSNR} = 10\log_{10}\frac{255^2}{\mathrm{MSE}},$$

$$H(R_t) = -\sum_b p_b \log_2 p_b,$$

where residual values are rounded to integers before estimating p_b.

1.2 Exhaustive Block Matching (35 pts)

Implementation Requirements: Implement:

```
def block_matching(prev_frame, cur_frame, block_size=16, search_radius=8,
    metric="sad"):
    ...
```

Divide the current frame into non-overlapping $B \times B$ blocks. For each current block, search displacements (u, v), $u, v \in [-R, R]$, in the previous frame and choose the best candidate by SAD or MSE. SAD means sum of absolute differences:

$$(u^\star, v^\star) = \arg\min_{u,v} \sum_{(x,y)\in\mathcal{B}} |I_t(x, y) - I_{t-1}(x + u, y + v)|\,.$$

1. Support `metric="sad"` and `metric="mse"`.
2. Return a motion-vector field of shape `(H // block_size, W // block_size, 2)`.
3. Store vectors as `(dy, dx)`, where the candidate block is sampled from `prev_frame[y + dy, x + dx]`.
4. Return the motion-compensated prediction $\hat{I}_t$ and residual $R_t = I_t - \hat{I}_t$.
5. Skip candidate blocks outside the image boundary.
6. Visualize motion vectors and compare against the baseline using MSE, PSNR, residual entropy, and runtime.

Evaluate $(\text{block\_size}, \text{search\_radius}) \in \{(8, 4), (16, 4), (16, 8), (32, 8)\}$.

---

2 Video Temporal Consistency Evaluation (30 pts)

Use at least 4 clips total from TA-provided videos or synthetic clips. If no external videos are available, use the clean-motion, flicker, jitter, and blur clips from `HW3.ipynb`.

2.1 Objective Metrics (15 pts)

Implement at least three of the following four metrics and report a table for every clip:

```
def frame_sharpness(clip): ...
def temporal_difference(clip): ...
def temporal_acceleration(clip): ...
```

1. **Frame sharpness:** average variance of Laplacian or average gradient magnitude.
2. **Temporal difference:** average $\|I_{t+1} - I_t\|_1$.
3. **Temporal acceleration:** average $\|I_{t+1} - 2I_t + I_{t-1}\|_1$, sensitive to flicker and jitter.
4. **Motion magnitude:** average block-motion magnitude from Section 1, or optical flow from an allowed library.

2.2 Human Evaluation (6 pts)

For each clip, give 1–5 ratings for visual quality, temporal consistency, and motion naturalness. Write one sentence per clip describing the most visible issue.

2.3 Evaluation Discussion (9 pts)

Answer briefly:

1. Which objective metric best matched your human judgment? Give one example.
2. Which objective metric failed or was misleading? Give one example.
3. Why is one scalar score usually insufficient for generated-video evaluation?

---

3 Short Conceptual Questions (20 pts)

Answer briefly. You may use equations, bullet points, or small tables.

3.1 Optical-Flow Constraint (10 pts)

Assume brightness constancy:

$$I(x+u, y+v, t+1) = I(x, y, t).$$

Using a first-order Taylor expansion, derive

$$I_x u + I_y v + I_t = 0.$$

Explain why the flow at a single pixel is under-constrained.

3.2 Space-Time Attention Complexity (10 pts)

For a latent video with T frames, $H \times W$ spatial tokens per frame, and hidden dimension d:

1. Derive the complexity of joint space-time self-attention over all THW tokens.
2. Derive the complexity of factorized attention: spatial attention within each frame, then temporal attention at each spatial location.
3. For $T = 16$, $H = W = 32$, explain why factorized attention is much cheaper.

---

4 Bonus (10 pts)

Choose one option.

- **Option A: Lucas-Kanade Optical Flow.** Implement `compute_image_gradients` and `lucas_kanade_flow`; visualize flow, warp the previous frame, and compare MSE/PSNR with block matching.
- **Option B: Hierarchical Block Matching.** Implement `build_gaussian_pyramid` and `hierarchical_block_matching`; compare runtime and PSNR against exhaustive search.
- **Option C: Temporal Self-Attention.** Implement a PyTorch temporal self-attention block for $x \in \mathbb{R}^{B \times T \times C \times H \times W}$. Verify shape and parameter count.

Suggested Materials

- Required course lectures: `L07_video_compression.pdf`, `L08_video_generation.pdf`.
- Optional course context: `L09_world_model.pdf`.
- Optional external references: MIT VisionBook Optical Flow and VBench.

Please submit your code, results, and report.